

Contents

Abstract	2
Background	3
Dataset Summary	3
Class definitions	3
Preprocessing	3
Data Splits	4
Exploratory Analysis	4
Method Description	4
Results	4
Exploratory Analysis and Model Performance	4
Model Interpretability	5
Conclusion & Discussion	5
Key Findings	5
Model Performance and Comparison	5
Model Interpretability	6
Limitations	6
Future directions	6
Data and Code Availability	6
Acknowledgments	6
References	7
Appendices	8
Figures	8
AI Transcripts	11
Deep Research Report	11

Abstract

This project developed a machine learning pipeline to classify 847 metabolites into five chemical groups using ^1H NMR spectroscopy data from the Human Metabolome Database. After binning spectra into 400 features (0.025 ppm resolution), three approaches were evaluated: two convolutional neural networks (15-23% accuracy) and Random Forest classifiers (66.4% accuracy). Feature importance analysis revealed chemically interpretable patterns: aliphatic regions (1.2-1.6 ppm) distinguished lipids, aromatic signals (6-8 ppm) identified aromatics, and alpha-to-heteroatom resonances (3.6 ppm) characterized nitrogen/oxygen-containing compounds. A Streamlit web application enables real-time classification with confidence scores, successfully validating on external spectra.

Background

Nuclear Magnetic Resonance (NMR) spectroscopy is an analytical technique that exploits the magnetic properties of atomic nuclei to study molecular structure. When placed in a strong magnetic field, nuclei with spin can absorb radiofrequency energy and resonate at characteristic frequencies. ^1H NMR is the most common type, since nearly all organic molecules contain hydrogen, but NMR can be used for any atomic nuclei with spin, such as ^{13}C , ^{19}F and ^{15}N . In ^1H NMR, each unique chemical environment around a hydrogen atom causes a slightly different resonance frequency (chemical shift), producing peaks in the spectrum. Peak splitting (multiplets) reveals information about neighboring atoms, while intensities reflect the number of protons, enabling detailed structural interpretation. ^1H NMR is used routinely for exact structure determination of small molecules in organic chemistry due to its speed and ease of use (typically ~ 5 min for a ^1H NMR measurement) but also has applications in metabolomics, protein and nucleic acid studies, material science and medical imaging (MRI).

Dataset Summary

^1H NMR data was retrieved from the Human Metabolome Database (HMDB) [1]. The HMDB data is made up of two parts: A .xml file containing various information for each compound in the database (HMDB ID, name and synonyms, InChI, description, molecular weight, CAS number, etc.), and a folder containing .txt files (see Code 1) with ^1H NMR peak lists (and in some cases also multiplets and assignments) for a subset of the compounds. (See Code 1)

Table of Peaks		
No.	(ppm)	Height
1	2.49	0.0970
2	2.50	0.1098
3	2.50	0.1134

Code 1: Structure of a typical ^1H NMR peaklist file. Three columns including peak number, chemical shift in ppm and peak height.

Class definitions

The original HMDB hierarchical classification system was consolidated to create five primary compound groups suitable for machine learning classification: Aromatics (127 samples, 15.0%), Heterocycles & nucleotides (180 samples, 21.3%), Lipids (189 samples, 22.3%), Nitrogenous & organic acids (251 samples, 29.6%), Organic oxygen compounds (100 samples, 11.8%).

Final dataset size: 847 compounds across 5 classes.

The class distribution shows moderate imbalance, with the largest class (Nitrogenous & organic acids) containing $2.5\times$ more samples than the smallest class (Organic oxygen compounds). This imbalance was addressed during model training through stratified sampling and class weighting.

Preprocessing

To convert variable-length peak lists into fixed-dimension feature vectors suitable for machine learning, spectra were binned using the following parameters: **ppm range** = 0-10 ppm. **Number of bins** = 400. **Resolution** = 0.025 ppm per bin. **Binning method** = Maximum intensity pooling (when multiple peaks fell within the same bin, the maximum intensity was retained). The binning resolution of 0.025 ppm was chosen to balance spectral detail with computational efficiency. This resolution is sufficient to distinguish most chemically meaningful peak differences while avoiding excessive sparsity.

A multi-step normalization procedure was applied to each binned spectrum. **Log transformation:** Applied $\log(1 + x)$ to compress the dynamic range of intensities, which can span several orders of magnitude. **Per-spectrum max normalization:** Each spectrum was scaled by dividing by its maximum bin value, normalizing to the $[0, 1]$ range. **Z-score normalization:** Features were standardized using mean and standard deviation calculated from the training set only, preventing data leakage.

This normalization pipeline ensures that the model learns from spectral patterns (peak positions and relative intensities) rather than absolute intensity scales, which can vary due to instrumental factors.

Data Splits

Data were partitioned into training, validation, and test sets using stratified random sampling to preserve class distributions: **Training set** = 592 samples (69.9%). **Validation set** = 127 samples (15.0%). **Test set** = 128 samples (15.1%). Stratification ensured each split maintained approximately the same class proportions as the full dataset. A fixed random seed (42) was used for reproducibility. The validation set was used for hyperparameter tuning and early stopping, while the test set remained completely held-out until final model evaluation.

Exploratory Analysis

Principal Component Analysis (PCA) revealed significant overlap between classes in low-dimensional projections. The first three principal components captured only 12.0% of total variance, indicating that discriminative information is distributed across many dimensions rather than concentrated in a few dominant features. This high-dimensional structure suggested that ensemble methods like Random Forest might outperform simpler linear classifiers or would require deep neural networks to capture complex non-linear decision boundaries.

Method Description

Workflow and models - Following data preprocessing and feature extraction (described in Data Summary), three modeling approaches were evaluated: two 1D Convolutional Neural Network architectures and Random Forest classifiers. The first CNN consisted of three convolutional blocks (32→64→128 filters, kernel size 5) with batch normalization, max pooling, dropout regularization (0.3-0.4), and dense classification layers (256→128 units), totaling 320,000 parameters. After poor performance, a simplified CNN was tested with only two convolutional blocks (16→32 filters, larger kernels of size 7 and 5) and a single 64-unit dense layer, reducing parameters by 98% to 5,157. Both CNNs were trained using Adam optimizer (learning rate 0.001), early stopping (patience 15-20 epochs), and class weighting to handle imbalance. The Random Forest baseline used 200 trees with max_depth=20, followed by systematic hyperparameter optimization via 5-fold cross-validated GridSearchCV across 648 configurations, testing n_estimators (100-500), max_depth (10-30/None), min_samples_split (2-10), min_samples_leaf (1-4), and max_features (sqrt/log2/None).

Evaluation metrics - Model performance was assessed using multiple metrics appropriate for multi-class classification. Primary metrics included accuracy (overall correct classification rate), weighted F1-score (harmonic mean of precision and recall, weighted by class support), and per-class precision and recall to identify class-specific strengths and weaknesses. Confusion matrices visualized misclassification patterns between classes. For the final tuned Random Forest model, feature importance analysis identified which ppm regions contributed most to classification decisions globally, while SHAP (SHapley Additive exPlanations) values provided per-prediction explanations showing how individual spectral features influenced specific classification outcomes. All models were evaluated on a held-out test set (15% of data) that remained untouched during training and hyperparameter tuning to provide unbiased performance estimates.

Results

Exploratory Analysis and Model Performance

Principal Component Analysis (Figure 1) revealed substantial overlap between compound classes, with the first three principal components capturing only PC1: 5.87%, PC2: 3.53%, and PC3: 2.65% of variance respectively. The scree plot shows a gradual increase in cumulative variance with no clear elbow, reaching approximately 30% with the first 20 components, indicating that discriminative information is highly distributed across spectral features. This lack of clear class separation in low-dimensional projections suggested that linear classification methods would struggle, motivating the evaluation of both ensemble and deep learning approaches.

Three modeling strategies were systematically compared. The initial CNN architecture (Figure 2) with 320 000 parameters achieved only 23.4% test accuracy, barely exceeding random chance (20%).

Training curves revealed severe overfitting, with validation loss exploding beyond 9000 while training loss remained near zero. The confusion matrix showed prediction collapse, with the model defaulting predominantly to a single class. A simplified CNN with 98% fewer parameters (Figure 3) showed 22.7% test accuracy but exhibited similar overfitting patterns with validation loss reaching 20 000, confirming that convolutional architectures were fundamentally unsuited to this sparse, high-dimensional data with limited samples.

In stark contrast, Random Forest methods substantially outperformed deep learning approaches. The baseline Random Forest achieved 64.8% test accuracy (Figure 4, top right), and systematic hyperparameter optimization via GridSearchCV across 648 configurations improved performance to 66.4% test accuracy, a modest but meaningful 1.6 percentage point gain. The confusion matrix (Figure 4, top left) revealed class-specific performance variations: Lipids exhibited the strongest diagonal (25/29 correct, 86% recall), while Heterocycles & nucleotides showed the lowest performance (9/27 correct, 33% recall), frequently confused with Nitrogenous & organic acids. Per-class accuracy analysis (Figure 4, bottom left) confirmed this pattern, with Lipids (86%) and Nitrogenous & organic acids (82%) performing best, and Heterocycles & nucleotides (33%) and Organic oxygen compounds (47%) performing worst. The cross-validation score distribution (Figure 4, bottom right) showed the optimal configuration achieved 69.3% CV accuracy with most parameter combinations clustering between 62-66%, suggesting the tuned model performance is near the achievable ceiling for this feature representation.

Model Interpretability

Feature importance analysis (Figure 5) demonstrated that the Random Forest learned chemically meaningful patterns. The most discriminative spectral region was 3.63 ppm (importance: 0.0170), corresponding to protons alpha to heteroatoms, characteristic of alcohols, ethers, and amino acids. The aliphatic region (1.2-1.6 ppm) dominated the top 20 features with multiple peaks (1.50, 1.60, 1.28, 1.30, 1.55, 1.53 ppm), reflecting the importance of methylene and methyl groups in distinguishing lipid structures. Aromatic signals (7.32, 6.87 ppm) also appeared prominently, indicating their role in classifying aromatic compounds. The NMR-style spectrum visualization revealed distinct importance profiles across the chemical shift range, with pronounced peaks in aliphatic (0.5-2 ppm), alpha-to-heteroatom (3-4 ppm), and aromatic (6-8 ppm) regions.

SHAP analysis (Figure 6) provided complementary insights into feature contributions across all classes. The summary plot confirmed that 3.63, 1.60, and 7.32 ppm were the most impactful features globally, with color-coded bars indicating differential effects across compound classes. Notably, aliphatic regions (blue, Lipids) dominated the contributions for lipid classification, while aromatic regions showed strong associations with Aromatics (orange). Per-class SHAP importance (Figure 7) revealed class-specific discriminative regions: Aromatics relied heavily on 7.32 and 6.92 ppm (aromatic protons), Lipids on 1.60 and 1.63 ppm (aliphatic chains), Nitrogenous & organic acids on 3.63 and 1.25 ppm (alpha carbons and backbone), and Organic oxygen compounds on 3.63 ppm (protons adjacent to oxygen). These patterns align with expected chemical functional groups.

Conclusion & Discussion

Key Findings

This project successfully developed a machine learning pipeline for classifying metabolite compounds based on ¹H NMR spectroscopy data from the Human Metabolome Database. The final Random Forest model achieved 66.4% test accuracy across five compound classes, demonstrating that automated classification of metabolites from spectral data is feasible, though challenging given the inherent complexity and overlap between chemical classes.

Model Performance and Comparison

Two CNN architectures were tested - a complex model (320 000 parameters) and a simplified version (5157 parameters). Both significantly underperformed, achieving only 23.4% and 22.7% test accuracy

respectively, barely above random chance (20%). The CNNs exhibited severe overfitting with validation loss exploding beyond 9000, indicating they struggled to learn generalizable patterns from the sparse, high-dimensional spectral features. Initial implementation of Random Forest classifier with 200 trees and `max_depth=20` achieved 64.8% test accuracy, substantially outperforming the CNNs. Systematic optimization improved performance to 66.4% test accuracy, representing a 1.6 percentage point gain over the baseline.

The dramatic superiority of Random Forest over deep learning methods was unexpected but aligns with the data characteristics: with only 847 samples, high feature sparsity, and substantial class overlap (as revealed by PCA), ensemble methods proved more suitable than neural networks requiring large training sets and dense representations.

Model Interpretability

Feature importance analysis revealed that the Random Forest model learned chemically meaningful patterns rather than spurious correlations. **Aliphatic region (1.2-1.6 ppm):** Most discriminative for Lipids, corresponding to methylene groups in fatty acid chains. **Alpha to heteroatom region (3.6 ppm):** Critical for distinguishing Nitrogenous & organic acids and Organic oxygen compounds, where protons adjacent to oxygen or nitrogen atoms resonate. **Aromatic region (6-8 ppm):** Important for Aromatics and Heterocycles & nucleotides, reflecting characteristic aromatic proton chemical shifts.

SHAP analysis demonstrated that individual classification decisions aligned with expected chemical functional groups. For instance, lipid classifications were driven primarily by signals in the 1.3-1.6 ppm range, while aromatic compounds showed strong contributions from the 6.9-7.3 ppm region.

Limitations

The primary limitation was the modest dataset size of 847 compounds, which proved insufficient for deep learning approaches and limited the model’s ability to capture rare spectral patterns. Class imbalance, with the largest class containing $2.5\times$ more samples than the smallest, contributed to lower performance on minority classes despite stratified sampling and class weighting.

The hierarchical grouping from HMDB’s original taxonomy into five super-classes involved subjective decisions, introducing classification ambiguity. Some compounds possess characteristics of multiple categories, establishing an inherent ceiling on achievable accuracy even for expert chemists.

Methodologically, converting variable-length peak lists into fixed 400-bin vectors discarded fine spectral details including peak multiplicity patterns. NMR spectra are also sensitive to experimental conditions (solvent, pH, temperature), introducing variability that complicates pattern recognition. The model’s 66.4% accuracy, while substantially better than CNNs, still misclassifies one-third of compounds and lacks explicit uncertainty quantification for flagging unreliable predictions.

Future directions

Future work should prioritize expanding the dataset through crowdsourcing or database partnerships, potentially reaching thousands of samples to enable more sophisticated modeling approaches.

Advanced architectures including graph neural networks (which encode molecular structure directly), hybrid CNN-ensemble methods, and transfer learning from large unlabeled spectral datasets could overcome current performance limitations. The web application could be enhanced with batch processing, confidence thresholding with automatic flagging of uncertain predictions, and integration with laboratory information systems for real-time deployment.

Data and Code Availability

All code and figures are publicly available in the [Github repo](#).

Acknowledgments

ChatGPT and Claude were used for assistance with this project. The author thanks the DDLS course teachers for an informative and fun course.

References

- [1] D. S. Wishart *et al.*, “HMDB 5.0: the Human Metabolome Database for 2022,” *Nucleic Acids Research*, vol. 50, no. D1, pp. D622–D631, 2022, doi: [10.1093/nar/gkab1062](https://doi.org/10.1093/nar/gkab1062).

Appendices

Figures

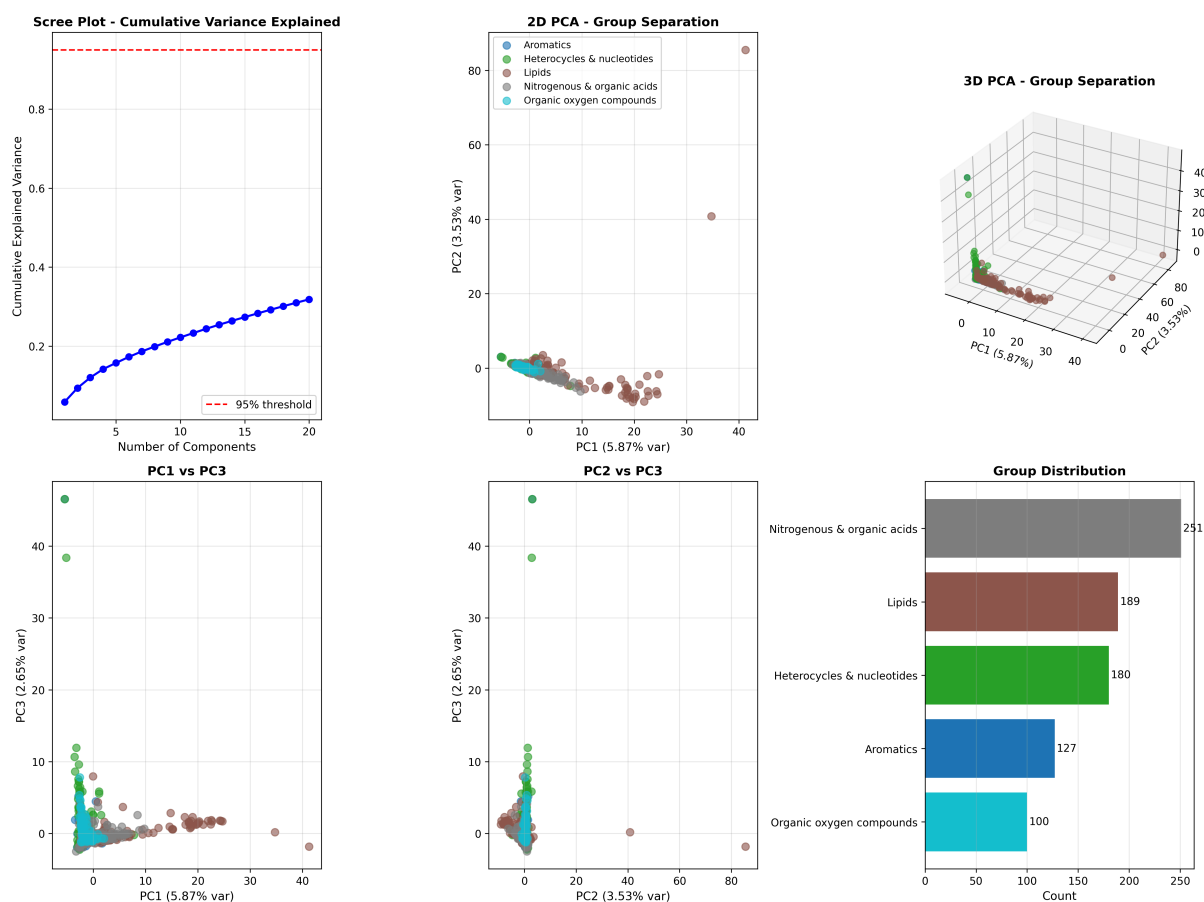


Figure 1: Principal Component Analysis of binned H-NMR spectral features.

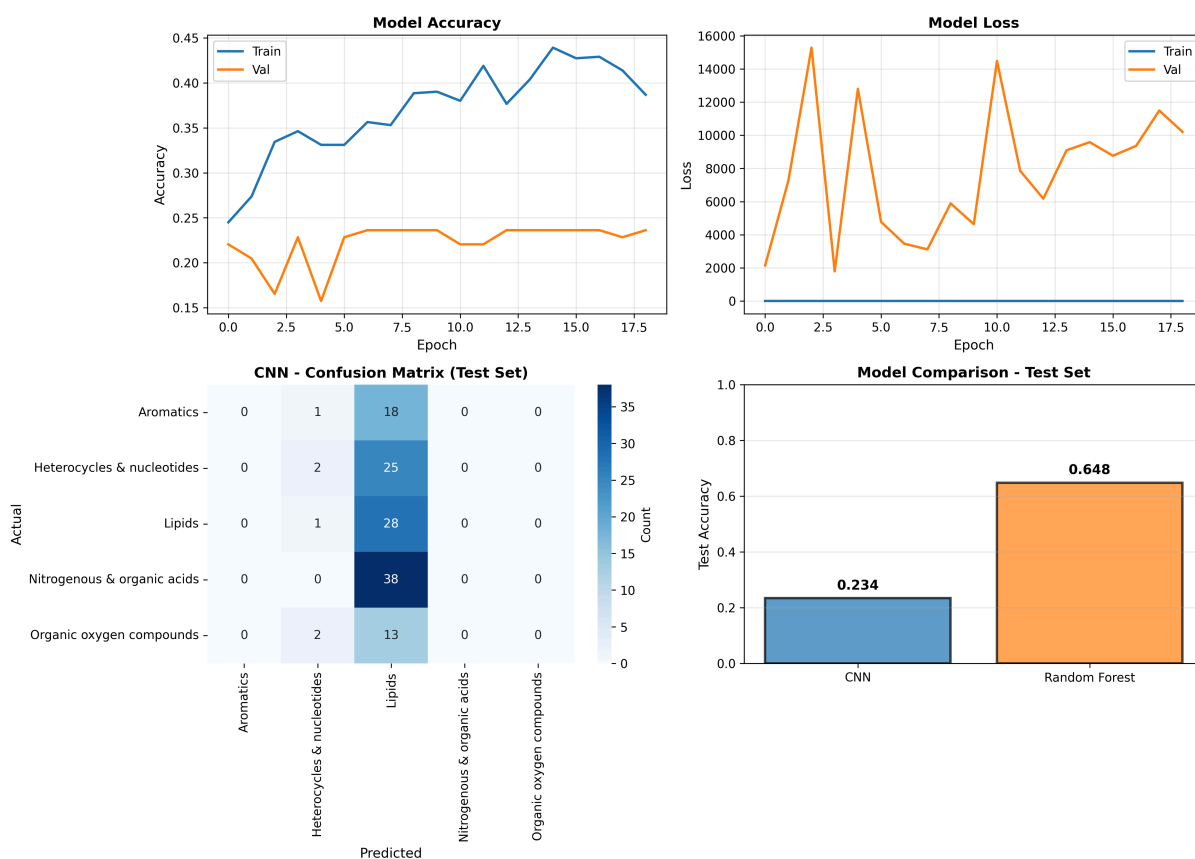


Figure 2: First CNN model training results.

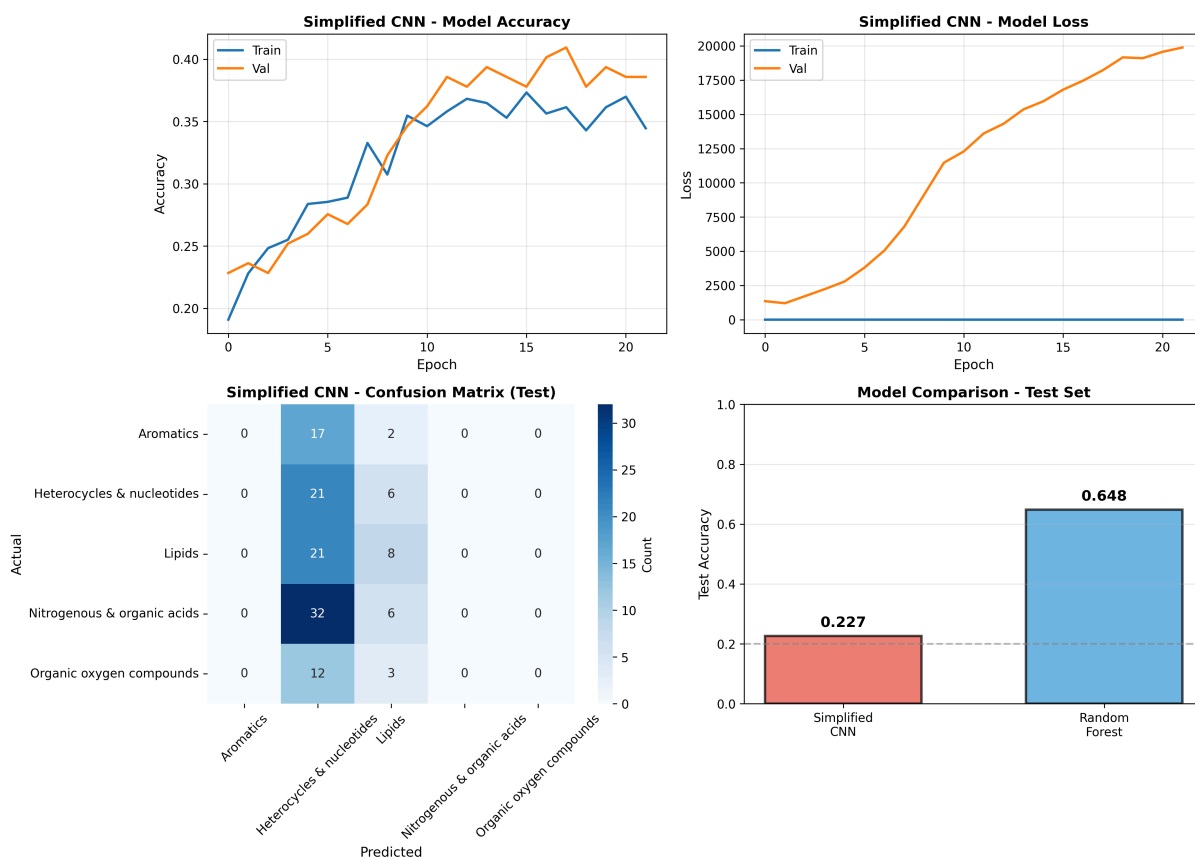


Figure 3: Simplified CNN model training results.

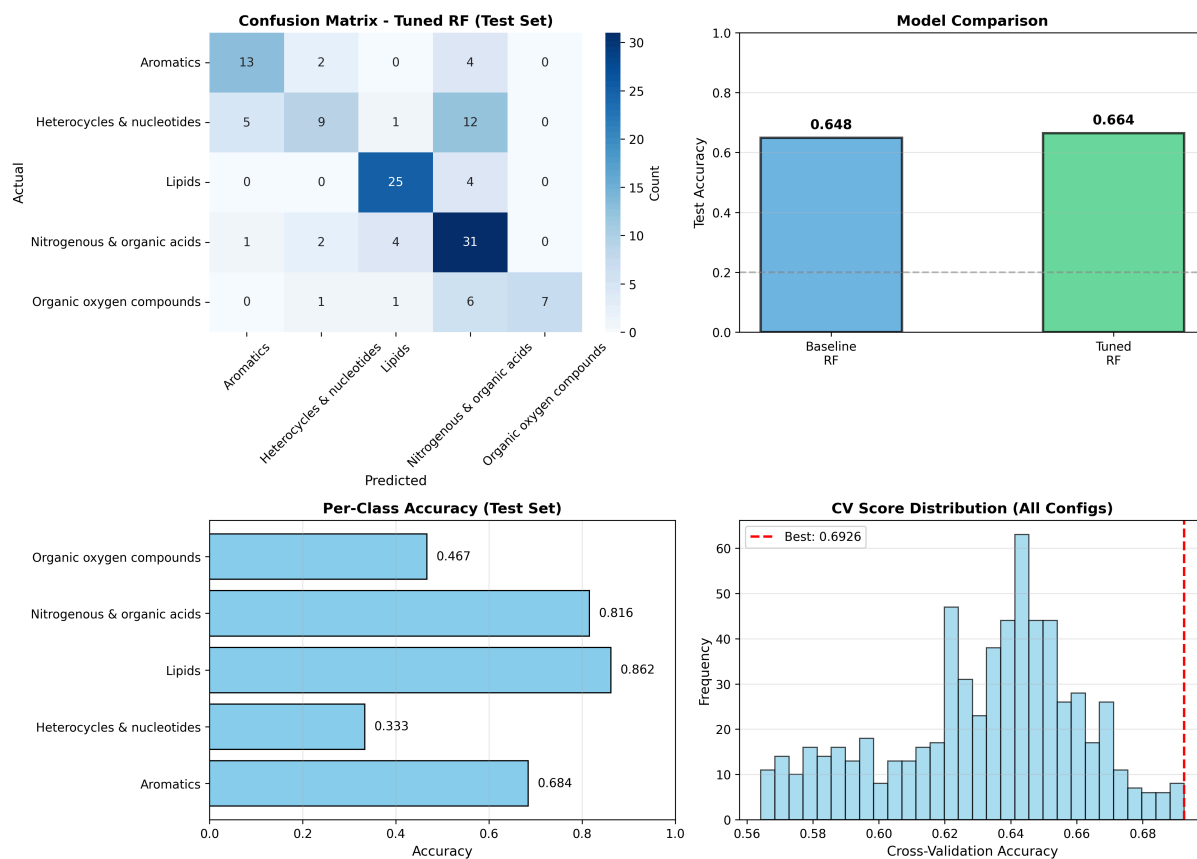


Figure 4: Tuned RF model results.

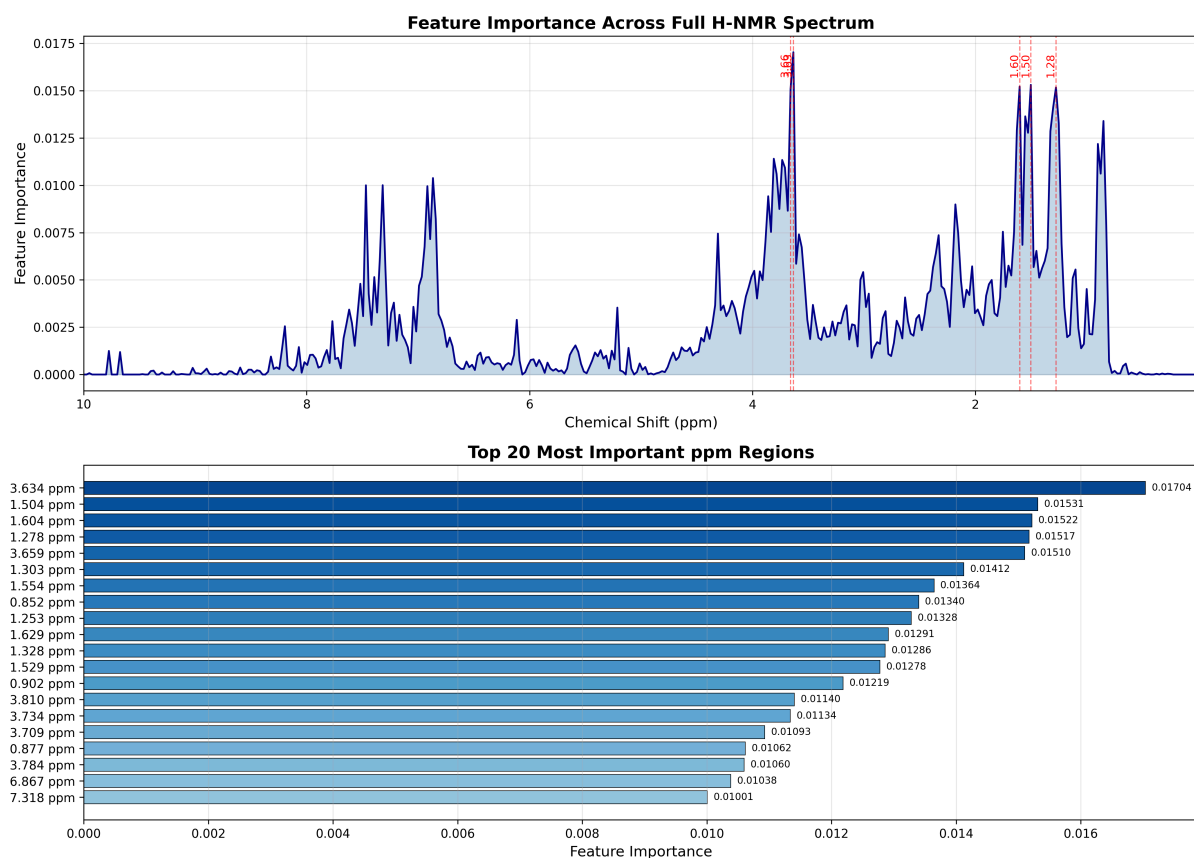


Figure 5: RF model feature importance.

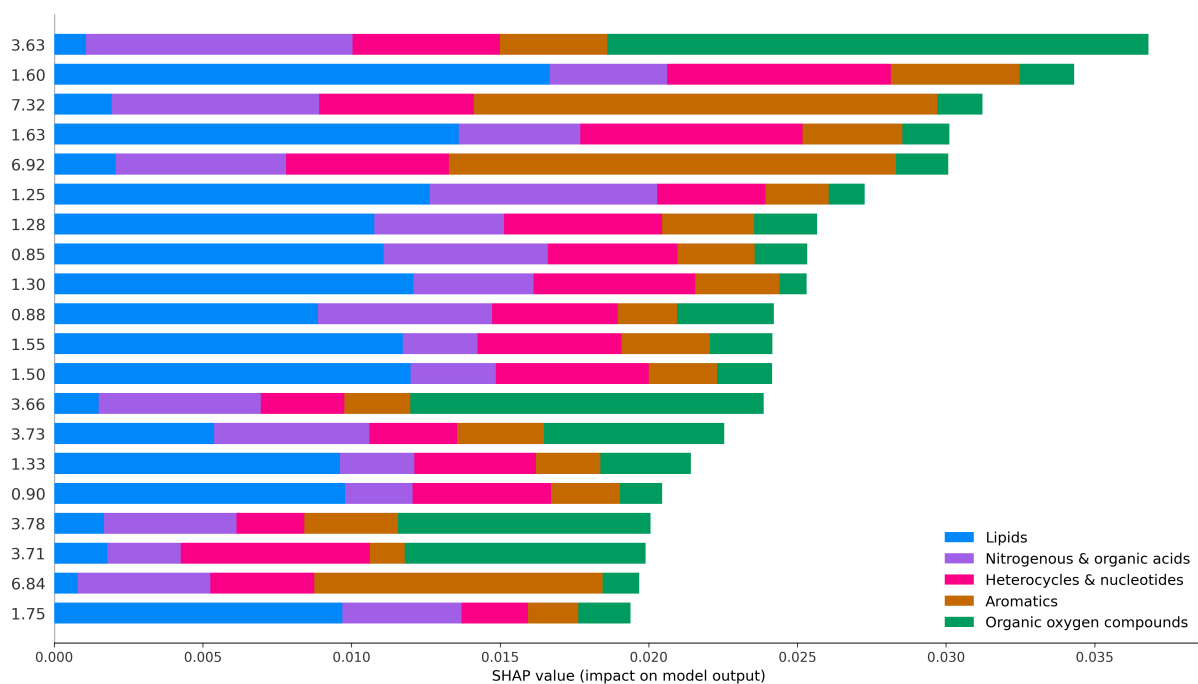


Figure 6: SHAP summary plot for RF model.

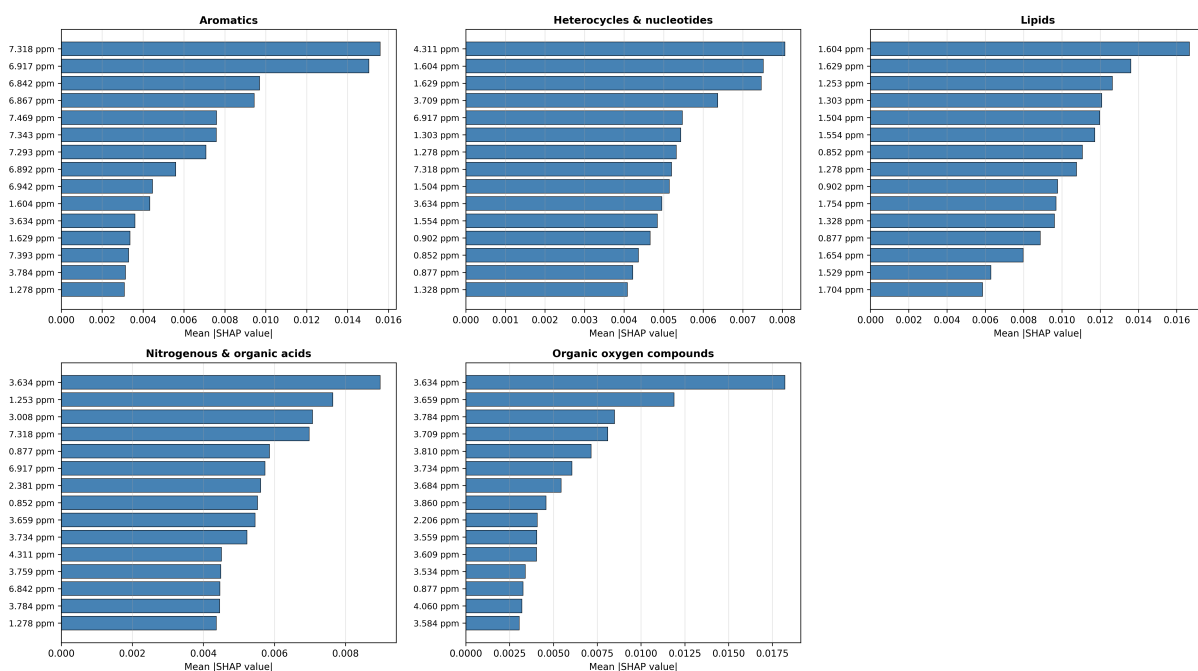


Figure 7: SHAP per class importance for RF model.

AI Transcripts

[Link to ChatGPT transcripts.](#)

[Link to Claude transcripts.](#)

Deep Research Report

Deep Research Report — ^1H -NMR Peak-List Classification

Project: DDLS 2025 Final Project — Alfred Larsson

Date: 2025-10-08

Agent: ChatGPT (GPT-5 Thinking)

1) Objective

Run a “Deep Research” to survey existing research, alternative approaches, and relevant data sources for classifying small molecules into metabolite families from **^1H -NMR peak-list spectra**. Provide concrete sources, data access points, and methodological options suitable for a 2–3 week project.

2) Key Findings at a Glance

- **Data sources:** HMDB (peak-list TXT, spectrum pages), BMRB (metabolomics entries, GISSMO library), nmrshiftdb2 (open database of assigned shifts & prediction), and COLMAR (query, ppm tools, HSQC databases).
 - **Methods used in literature:** Random Forest, SVM/PLS-DA/OPLS-DA baselines; CNNs and autoencoders for 1D spectra; newer **GNNs/transformers** for shift prediction and spectral learning.
 - **Augmentation:** **Predict chemical shifts** from structures (COLMARppm, PROSPRE, nmrshiftdb2) to expand training data and **normalize by solvent/field**.
 - **Feasibility:** HMDB + BMRB provide enough compounds to build a multi-class classifier across superclasses (Amino acids, Sugars, Lipids, Nucleotides, Organic acids) within the course constraints.
-

3) Datasets & Access Points

3.1 Human Metabolome Database (HMDB)

- **Peak-list bundle:** “Raw NMR Spectra Peaklist Files (TXT)” (contains ppm/intensity lists).
- **Per-spectrum pages** provide *List of chemical shift values (TXT)* and **nmrML/JDX**.
- **License/terms:** Free to use with acknowledgement; commercial use requires permission.
- **Notes for this project:** HMDB TXT files carry HMDB Accessions; match to XML metadata for **Class/Superclass** labels.

Links:

- Downloads index (shows “Raw NMR Spectra Peaklist Files (TXT)”).

- Example spectrum pages with **TXT peak lists** in water at 500 MHz.

3.2 BMRB (Biological Magnetic Resonance Data Bank)

- **Metabolomics portal** with bulk downloads (entries tarball, relational CSV/SQL).
- **GISSMO** library provides curated small-molecule spectra; downloadable in **NMR-STAR** and **NMReDATA** formats.
- **Use in project:** Supplement HMDB with additional compounds; harmonize to **peak lists** via vendor tools or nmrML parsers if needed.

3.3 nmrshiftdb2

- Open database supporting **assigned shifts** and **prediction** ($^1\text{H}/^{13}\text{C}$).
- **Use in project:** Generate **predicted ^1H peak lists** (solvent/field aware if available) to **augment** minority classes or to pretrain an **autoencoder**.

3.4 COLMAR / COLMARppm

- Web tools and databases (incl. HSQC for mixtures) and **COLMARppm** for **rapid $^1\text{H}/^{13}\text{C}$ shift prediction** and automated assignment.
- **Use in project:** Benchmark or augment with predicted shifts; check ppm tolerances and solvent conditions.

4) Methods Landscape & Alternative Approaches

4.1 Supervised Baselines (Shallow)

- **RF / SVM / Logistic / PLS-DA** on binned ^1H spectra (e.g., 0.01–0.04 ppm bins).
- **Pros:** Fast, interpretable feature importances, aligns with your current pipeline.
- **Refs:** Evaluation studies comparing multivariate classifiers on simulated/experimental ^1H -NMR; themed reviews on ML in NMR.

4.2 Deep Learning for 1D ^1H Spectra

- **1D-CNN** for sequence-like spectra; **denoising autoencoders** $\rightarrow$ classifier.
- **Spectrum-aware models:** e.g., **NMRQNet** for identification/quantification from NMR spectra.
- **Transformers / contrastive:** pretrain on predicted/real spectra (SimCLR-style) to learn ppm-invariant embeddings.

4.3 Graph-Based Chemical-Shift Predictors (for Augmentation)

- **GNN-based** chemical-shift prediction from small data; **PROSPRE** (deep learning, solvent-aware ^1H shift prediction).
- **Augmentation idea:** From **SMILES** $\rightarrow$ predicted **^1H shifts** $\rightarrow$ synthesize peak lists (add line-shapes if needed), then train classifier; helpful for **class imbalance**.

4.4 Domain-Specific Tools

- **COLMARppm** for fast shift predictions and assignments (can generate clean synthetic peaks).
 - **GISSMO** curated lineshapes aid realistic synthetic spectra (Lorentzian/Gaussian).
-

5) Practical Recommendations for This Project

1. **Core dataset:** HMDB peak-list TXT + matched **Superclass/Class** from HMDB XML (stream parse).
 2. **Supplement:** Add BMRB/GISSMO entries where classes are under-represented.
 3. **Preprocessing:**
 - **Solvent/field normalization:** keep water/D2O and 500 MHz subsets; otherwise apply **ppm alignment** and **intensity scaling** (unit max or total integral).
 - **Binning:** 0.02 ppm bins in [-1, 14] ppm is a good start; test 0.01 and 0.04.
 - **Peak to vector:** convert lists to binned intensities; optionally **peak-picking** → **sparse vector**.
 4. **Baselines:** RF, Logistic, and **PLS-DA** with 5-fold, report **balanced accuracy & weighted F1**.
 5. **Improvements:**
 - **1D-CNN** on normalized vectors; small architecture (100k params).
 - **Autoencoder pretraining** or **contrastive pretraining** (real+predicted spectra).
 - **Class weights** or **SMOTE-like augmentation** with **predicted shifts** for minority classes.
 6. **Error analysis:** Confusion matrix by **Superclass**; inspect ppm regions driving misclassifications (use permutation/SHAP on bins).
 7. **Accessibility:** Minimal **Streamlit** app: upload peak list → binned vector → prediction + top-bins explanation.
-

6) Risks & Mitigations

- **Instrument/solvent heterogeneity:** Restrict to consistent conditions (e.g., H₂O/D₂O, 500 MHz) or apply alignment; document solvent/field in features.
 - **Class imbalance:** Use **class weights** and **shift-prediction augmentation** (COLMARppm/PROSPRE).
 - **XML size (HMDB): Streaming parse** to extract only accession + class fields.
-

7) Citations & Source Pointers

- **HMDB Downloads** (lists “Raw NMR Spectra Peaklist Files (TXT)”: *Human Metabolome Database — Downloads*).

- Example HMDB spectrum pages with **TXT lists** and nmrML/JDX: examples for 500 MHz H¹ O.
- **BMRB metabolomics** bulk downloads and relational DB; **GISSMO** library page.
- **nmrshiftdb2** overview & usage; GNN shift-prediction on nmrshiftdb2 data (Rull 2023).
- **COLMARppm** (Rigel 2023/2024) — ACS Anal. Chem. article & PubMed; free PMC article.
- **NMRQNet** (Wang 2023) for deep NMR analysis.
- **Evaluation of multivariate classifiers** for 1D ¹H-NMR (Vu 2019).
- **Tea origin classification** via ¹H-NMR + ML (Hou 2024) — an applied example.
- Review: **Machine learning in NMR spectroscopy** (Klukowski 2025).
- Perspective on metabolomics & COLMAR databases (Markley 2017).

(Full URLs and metadata embedded in the accompanying transcript below.)

8) Search Transcript (Prompts & Results)

Prompt 1: “HMDB NMR peaklist files proton NMR peaks TXT download HMDB”

Result: HMDB downloads page lists “Raw NMR Spectra Peaklist Files (TXT)”.

Prompt 2: “classification of metabolites from 1H NMR spectra machine learning study”

Results: NMRQNet (2023); classifier comparisons for 1D ¹H-NMR (2019); applied studies (e.g., 2024 tea origin).

Prompt 3: “BMRB small molecule NMR database metabolomics peak lists download”

Results: BMRB metabolomics portal; bulk downloads; GISSMO library.

Prompt 4: “nmrshiftdb2 dataset download predicted 1H NMR chemical shifts small molecules”

Results: nmrshiftdb2 site, documentation; shift-prediction GNN paper (2023).

Prompt 5: “COLMAR NMR database small molecules 1H 13C metabolomics”

Results: COLMARppm articles (2023/2024) describing fast ¹H/¹³C shift prediction and assignment.

9) How to Cite in Your Plan

- Cite HMDB downloads and specific spectrum pages you used (include accession IDs).
- If using BMRB/GISSMO, cite the metabolomics portal and GISSMO.

- If augmenting with predicted shifts, cite **COLMARppm** and/or **PROSPRE** with solvent details.
-

10) Appendix: Direct Links

- HMDB Downloads — “Raw NMR Spectra Peaklist Files (TXT)”.
 - HMDB per-spectrum examples (TXT lists, 500 MHz H\ O).
 - BMRB metabolomics portal; bulk downloads.
 - GISSMO library.
 - nmrshiftdb2 home; usage guide; Rull 2023 GNN shift prediction.
 - COLMARppm (ACS Anal. Chem. paper; PubMed; PMC open text).
 - NMRQNet (PMC).
 - Classifier evaluation for 1D ¹H-NMR (PMC).
 - Machine learning in NMR spectroscopy (2025 review).
 - Metabolomics perspective incl. COLMAR (2017).
 - Applied example (tea origin classification, 2024).
-

Prepared for inclusion as the “Deep Research transcript/report” in the DDLS Step-1 Project Plan.